

CBSE Practice Worksheet

Class 7 | Science | Difficulty: Hard | Set D

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. What is the escape velocity from Earth's surface?
(A) 8 km/s
(B) 11.2 km/s
(C) 15 km/s
(D) 25 km/s
2. Which type of mutation involves change in single nucleotide?
(A) Deletion
(B) Insertion
(C) Point mutation
(D) Frameshift
3. What is the hybridization of carbon in methane?
(A) sp
(B) sp^2
(C) sp^3
(D) sp^3d
4. Which principle explains the working of a hydraulic lift?
(A) Archimedes
(B) Pascal
(C) Bernoulli
(D) Newton
5. What is the efficiency of an ideal Carnot engine operating between 300K and 400K?
(A) 25%
(B) 33%
(C) 50%
(D) 75%
6. What is the magnetic quantum number for 3d orbital?
(A) -2 to +2
(B) -3 to +3
(C) -1 to +1
(D) 0 only
7. Which quantum number determines the shape of an orbital?
(A) Principal
(B) Azimuthal
(C) Magnetic
(D) Spin
8. Which organelle is involved in programmed cell death?
(A) Lysosome
(B) Mitochondria
(C) Peroxisome

(D) Ribosome

9. What is the coordination number in FCC structure?

- (A) 6
- (B) 8
- (C) 12
- (D) 14

10. Which reaction is endothermic?

- (A) Combustion
- (B) Neutralization
- (C) Photosynthesis
- (D) Respiration

Section B: Fill in the Blanks (1 mark each)

- 11. The process of splitting water during photosynthesis is called ____.
- 12. The Krebs cycle occurs in the ____ of mitochondria.
- 13. The magnetic field inside a solenoid is given by $B = \mu \blacksquare$ ____.
- 14. The IUPAC name of $\text{CH}_3\text{CH}_2\text{CHO}$ is ____.
- 15. For a reversible process, entropy change of the universe is ____.
- 16. In $\text{S}_\text{N}2$ reactions, the rate depends on both ____ and nucleophile.
- 17. The uncertainty principle was given by ____.
- 18. Raoult's law applies to ____ solutions.

Section C: Short Answer Questions (2 marks each)

- 19. Explain Le Chatelier's principle with examples.
- 20. Describe the light-dependent and light-independent reactions of photosynthesis.
- 21. Derive the equation for kinetic energy of gases from kinetic theory.
- 22. Derive the lens formula and explain its applications.
- 23. Describe the mechanism of nerve impulse transmission.
- 24. Derive the expression for electric field due to a dipole.
- 25. Explain hybridization with examples of sp , sp^2 , and sp^3 .

--- End of Worksheet ---

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